Arista Access Node HSR-ACC-69 Hardware/Environmental Fault Resolution SOP

Node: HSR-ACC-69 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_3870

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve a critical hardware/environmental fault on Arista Access Node HSR-ACC-69. The failure domain is characterized by ASIC memory parity errors, line card crashes, and power supply redundancy loss. The root cause of this failure domain is attributed to the complex interplay between the Arista EOS (Extensible Operating System) and the underlying hardware components. Specifically, the EOS relies on the ASIC (Application-Specific Integrated Circuit) to handle packet processing, forwarding, and switching. However, the ASIC's memory parity errors can lead to catastrophic failures, including line card crashes and power supply redundancy loss. Furthermore, the EOS's reliance on the ASIC's hardware components can exacerbate the issue, making it challenging to diagnose and resolve. Additionally, the HSR-ACC-69 node's role as an access node, which provides connectivity to edge devices and networks, makes it a critical component of the overall network architecture. Therefore, prompt resolution of this failure domain is essential to ensure network uptime and prevent service disruptions.

Pre-requisite Checks

- pre_requisite: Ensure remote hands are dispatched to HSR Layout Access for physical access and assistance., description: This is a critical prerequisite to ensure safe and efficient resolution of the failure domain.

- pre_requisite: Verify the status of redundant links to ensure connectivity to downstream devices and networks., description: This step is essential to prevent network disruptions and ensure that the resolution process does not exacerbate the issue.

- pre_requisite: Check the power supply redundancy status to ensure that the node is not operating in a degraded mode., description: This step is critical to prevent further hardware damage and ensure safe resolution of the failure domain.

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system internal hasic memory, Expected Output: memory information, Description: This command will display detailed
```

```
information about the ASIC's memory, including parity errors and
inconsistencies. This will help identify the root cause of the failure
domain.

Command: show system internal hasic hardware, Expected Output:
hardware information, Description: This command will display detailed
information about the ASIC's hardware components, including
temperature, voltage, and other critical metrics. This will help
identify any hardware-related issues that may be contributing to the
failure domain.

Command: show system internal hasic status, Expected Output: status
information, Description: This command will display detailed
information about the ASIC's status, including any errors, warnings,
or critical issues. This will help identify the root cause of the
failure domain and guide the resolution process.
```

Resolution Steps

Step 1: Step: Replace the failed ASIC, Command: replace system internal hasic, Description: This step will replace the failed ASIC with a new one, which will resolve the hardware-related issues contributing to the failure domain. Ensure that the new ASIC is properly configured and validated before proceeding.

Step 2: Step: Replace the failed line card, Command: replace system internal line-card, Description: This step will replace the failed line card with a new one, which will resolve any line card-related issues contributing to the failure domain. Ensure that the new line card is properly configured and validated before proceeding.

Step 3: Step: Restore power supply redundancy, Command: enable power-supply redundancy, Description: This step will restore power supply redundancy to the node, which will prevent further hardware damage and ensure safe resolution of the failure domain.

Post-Check Verification

```
Command: show system internal hasic memory, Expected Output: memory
information, Description: This command will verify that the ASIC's
memory is functioning correctly and that parity errors have been
resolved.

Command: show system internal hasic hardware, Expected Output:
hardware information, Description: This command will verify that the
ASIC's hardware components are functioning correctly and that any
issues have been resolved.

Command: show system internal hasic status, Expected Output: status
information, Description: This command will verify that the ASIC's
status is normal and that any errors, warnings, or critical issues
have been resolved.
```

Roll-back Procedures

- Step: Restore the original ASIC, Command: restore system internal hasic, Description: This step will restore the original ASIC, which will revert the node to its original state. Ensure that the original ASIC is properly validated and configured before proceeding.

- Step: Restore the original line card, Command: restore system internal line-card, Description: This step will restore the original line card, which will revert the node to its original state. Ensure that the original line card is properly validated and configured before proceeding.

- Step: Disable power supply redundancy, Command: disable power-supply redundancy, Description: This step will disable power supply redundancy, which will revert the node to its original state. Ensure that the node is properly validated and configured before proceeding.